Teacher Guide: Week 1

Logic Is Not Magic: What Logic Can and Cannot Do

12th Grade Long Logic Unit Based on Richard Whately's *Elements of Logic*

Week 1 Overview

This first week introduces Richard Whately's basic account of logic for a twelfth-grade long unit. Students should leave the week understanding that logic is not a machine that automatically discovers truth. Logic studies the structure of reasoning and helps expose bad inferences. The reading teaches the pattern of conclusion, stated reason, hidden assumption, and test; the argument lab then asks students to transfer that pattern to fresh examples and to the recurring Shakespeare anchors, *Macbeth* and *Julius Caesar*. This establishes the foundation for later weeks on terms, propositions, syllogisms, mood and figure, fallacies, ambiguity, induction, rhetoric, and verbal versus real disputes.

Essential Question

What can logic actually do for a thinker, and what should we not expect it to do?

Student-Facing Materials

- *Logic Is Not Magic: What Logic Can and Cannot Do* — student reading handout.
- *Week 1 Argument Lab: What Follows From What?* — transfer-practice handout with fresh argument tests, a repair task, an original argument task, and an exit claim.
- *Macbeth Lit-Anchor Reading: Prophecy Is Not Proof* — Shakespeare anchor reading with both Week 1 Macbeth passages: Act 1, Scene 3 and Act 1, Scene 7.
- *Macbeth Anchor Worksheet* (U1L1AW) — AI-evaluable worksheet using scope-style Part/Q prompts and ANSWER BOX labels for applying Whately's four questions to Macbeth's movement from prophecy to inference, ambition, and moral responsibility.
- *Week 1 Logic Test: The Case of the Three Bells* — short puzzle-style assessment requiring students to diagnose conclusion, stated reason, hidden assumption, what follows, and what logic cannot decide alone.

Learning Goals

Students will be able to:

1. Define logic as the study of reasoning and argument structure.

2. Distinguish what logic can do from what it cannot do.
3. Identify conclusion, stated reason, and hidden assumption in a simple argument.
4. Explain why popularity, public honor, tradition, difficult language, confidence, or emotion do not by themselves prove a claim.
5. Repair a weak argument by narrowing the claim or strengthening the support.
6. Create and test an original argument.
7. Apply the four-step test to an initial *Macbeth* or *Julius Caesar* claim for the course portfolio.
8. Transfer Whately's four-question test to a puzzle scenario and distinguish logical inference from missing facts, testimony, and moral judgment.

Key Vocabulary

logic, reasoning, argument, premise, conclusion, deduction, fallacy, science, art, panacea, hidden assumption

Weekly Lesson Sequence

Day 1: Opening Question and First Read

- Begin with the question: *If someone is good at arguing, does that mean he is good at finding truth?*
- Have students answer briefly in writing.
- Introduce Whately as a nineteenth-century thinker trying to rescue logic from both overpraise and contempt.
- Read the introduction and first half of the student reading aloud or in pairs.
- Stop after the analogy paragraph. Ask: What does Whately think logic is being unfairly expected to do?

Day 2: Finish Reading and Annotate Whately's Claim

- Finish the reading.
- Students mark three places: one sentence showing what logic can do, one showing what it cannot do, and one example that separates conclusion, stated reason, hidden assumption, and test.
- Complete Reading Focus questions 1–6.

Day 3: Argument Skeleton Practice

- Introduce the pattern: conclusion, stated reason, hidden assumption, test.
- Work through the modeled popularity/fairness example from the Argument Lab handout. Name it explicitly as the one repeated example from the reading.
- Students complete the fresh Part II arguments individually, then compare answers with a partner.
- Tell students that the goal is transfer: the reading already showed the method, but the lab asks whether they can use it on new arguments.
- Emphasize that Whately's basic question is: *What follows from what?*

Day 4: Macbeth Lit-Anchor: Prophecy Is Not Proof

- Read the Act 1, Scene 3 excerpt from the lit-anchor reading: the witches' greeting, Ross's confirmation of Cawdor, Banquo's warning, and Macbeth's first aside.
- Build a four-question skeleton on the board: What conclusion does Macbeth begin to draw? What reason does he have? What hidden assumption links prophecy to action? Does the conclusion follow?
- Keep the key distinction visible: the witches give a prediction, not a command. Macbeth adds interpretation, desire, and possible action.
- Have students read the Act 1, Scene 7 passage and begin the U1L1AW *Macbeth Anchor Worksheet*.

Day 5: Repair, Portfolio, and Exit Claim

- Students finish either the repaired argument/original test-case work from the lab or the U1L1AW *Macbeth Anchor Worksheet*.
- Require one reference to Whately's view and one example from school, literature, public speech, everyday life, or *Macbeth*.
- Use Macbeth to synthesize the week: logic can expose a bad inference, but it cannot make Macbeth virtuous.
- Give the short puzzle-style logic test, *The Case of the Three Bells*, as the week's assessment or as a take-home/open-note transfer task.
- Collect the student reading, argument lab, U1L1AW anchor worksheet, and logic test as the week's work product.

Lit-Anchor Teaching Notes

The Week 1 lit-anchor is now split into a reading handout and an AI-evaluable anchor worksheet. The reading handout keeps the Shakespeare passages, purpose, central question, and light side glosses. The U1L1AW worksheet carries the written work using stable Part/Q labels and scope-style ANSWER BOX labels. This is not a plot worksheet. Its purpose is to let students use Whately's modest account of logic on a familiar literary problem: Macbeth hears a prophecy, then begins making inferences from it. Keep students from saying simply, "the witches made him do it." The more precise Whately-style question is whether Macbeth's conclusions follow from the reasons available to him.

- **Guided passage:** *Macbeth* Act 1, Scene 3, from the witches' greeting through Macbeth's "If chance will have me king" aside.
- **Second passage:** Act 1, Scene 7, Macbeth's soliloquy before Duncan's murder.
- **Central logical pressure:** prediction is not permission; possibility is not command; desire is not proof.
- **Worksheet artifact:** U1L1AW Macbeth anchor worksheet: prophecy-versus-command, Macbeth's "happy prologues" inference, Banquo's rival interpretation, Act 1, Scene 7 reasons against murder, and a final Whately synthesis.

A strong class discussion should separate three claims: (1) Macbeth may become king; (2) Macbeth should act to become king; (3) murder may be a legitimate path to kingship. The first has some support once Cawdor is confirmed. The second and third outrun the evidence. Banquo's warning is important because it offers a rival interpretation: truthful fragments can still be bait.

Suggested Board Notes

- Logic does not discover every truth.
- Logic studies whether a conclusion follows from premises.
- Bad facts can still produce bad conclusions even when the reasoning looks neat.
- A reading example may teach the method; fresh examples test whether students can transfer it.
- Popularity, public honor, tradition, difficult language, confidence, and repetition are not the same as proof.
- A sign or rule may give evidence, but logic alone cannot prove that the sign was used correctly.
- The first logical question: What follows from what?

Argument Lab Structure

- **Part I** is a model from the reading. Do not treat it as the main challenge; use it to rehearse the four-part method.
- **Part II** contains four fresh arguments. Arguments 1–3 are deliberately weak transfer cases. Argument 4 is stronger, but only if the two students' situations are truly alike in all relevant ways.
- **Part III** asks students to repair one weak argument. Accept either a more careful conclusion or added support that actually bears on the conclusion.
- **Part IV** asks students to create an original argument and test their own reasoning.
- The exit claim asks students to explain when logic helps a reader most, not merely to praise logic in general.

Logic Test: The Case of the Three Bells

The Week 1 test is intentionally puzzle-like rather than a vocabulary quiz. It uses a small invented academy scenario to make students transfer Whately's four-question method to new material. The three bell claims test three errors already named in the reading and lab: applause mistaken for truth, confidence mistaken for proof, and a public sign or office mistaken for guaranteed fact.

Use the test after the argument lab and Macbeth lit-anchor. Students may complete it in one class period. If used open-note, require them to quote or paraphrase Whately's claim that logic tests reasoning but does not supply missing facts or virtue.

Logic Test Answer Guide

- **Clara / Iron Bell.** Conclusion: her argument must have been true. Stated reason: everyone applauded the speech. Hidden assumption: applause or popularity proves truth. Test: the conclusion does not follow. Applause may show approval, excitement, emotional force, or popularity, but it does not by itself prove the argument true.
- **Miles / Brass Bell.** Conclusion: he had the strongest proof. Stated reason: he spoke more confidently than anyone else. Hidden assumption: confidence proves strength of reasoning. Test: the conclusion does not follow. A confident speaker may be wrong, and a hesitant speaker may have strong evidence.
- **Jonas / Silver Bell.** Conclusion: he spoke wisely. Stated reason: the Silver Bell is supposed to be rung for wisdom, and he says he rang it. Hidden assumption: the bell was rung correctly, honestly, and according to its rule. Test: this is stronger than Clara's or Miles's reasoning because the bell's stated purpose is relevant, but it is still incomplete proof. The class still needs facts or testimony about who rang the bell and why.
- **Ranking.** Clara and Miles are both weak because they confuse signs of persuasion with proof. Jonas is the strongest or least weak because the Silver Bell's rule is directly related to wisdom, but the rule cannot prove that the event actually happened correctly.

- **What logic can decide.** Logic can test whether applause proves truth, whether confidence proves proof, and whether a conclusion follows from a stated reason. Logic alone cannot decide whether Jonas actually rang the Silver Bell honestly or whether a student is wise or virtuous in character; those require evidence, testimony, observation, and moral judgment.
- **Macbeth bridge.** Strong answers should say that Macbeth is tempted to treat prophecy, one fulfilled prediction, or imagined kingship as if it proves permission, certainty, or moral justification. Students should use conclusion, reason, hidden assumption, and follows accurately.

Reading Focus Answer Guide

1. Main claim: Logic is the science and art of reasoning; it studies argument structure and helps guard against bad deductions.
2. Can do: examine whether conclusions follow; expose skipped steps, changed terms, hidden assumptions, or bad inferences.
3. Cannot do: supply facts by itself; make foolish people wise; discover every truth; replace observation, experience, memory, imagination, or moral judgment.
4. Four steps: find the conclusion, find the stated reason, identify the hidden assumption, and test whether the conclusion really follows.
5. Answers will vary. Strong responses should use one reading example and correctly name its conclusion, reason, assumption, and test. For example, the Macbeth argument concludes that Macbeth is not responsible; its stated reason is the witches' prediction; its hidden assumption is that prediction removes responsibility; the conclusion does not necessarily follow. In the lit-anchor, students may also test Macbeth's more immediate inference that one fulfilled prediction makes kingship certain or that a predicted future authorizes action.
6. People blamed logic because some defenders promised too much for it, so critics rejected it when it failed to perform impossible tasks.

Argument Lab Answer Guide

The revised lab keeps one modeled argument from the reading, then moves students into fresh test cases. Strong student answers should not merely label parts; they should explain whether the reason actually earns the conclusion.

Model From the Reading: Popularity and Fairness

- Conclusion: The rule must be fair.
- Stated reason: Most students like it.
- Hidden assumption: Whatever most students like is fair.
- Test: The conclusion does not necessarily follow. Popularity may sometimes accompany fairness, but popularity alone does not prove fairness.

Argument 1: Statue and Noble Life

- Conclusion: The man must have lived nobly.
- Stated reason: The city placed a bronze statue of him in the town square.
- Hidden assumption: Public honor always goes to people who lived nobly, or a statue proves virtue.
- Test: The conclusion does not necessarily follow. A statue proves that a community chose to honor someone; it does not by itself prove that the person deserved the honor.

Argument 2: Difficult Words and Better Reasoning

- Conclusion: The first essay must have better reasoning.
- Stated reason: The first essay uses more difficult words than the second essay.
- Hidden assumption: More difficult language means better reasoning.
- Test: The conclusion does not follow. Difficult words may hide weak reasoning, and plain words may express strong reasoning.

Argument 3: Always Done This Way

- Conclusion: The old method must be the best method.
- Stated reason: The class has always done the assignment this way.
- Hidden assumption: Whatever has been done for a long time is best.
- Test: The conclusion does not necessarily follow. Tradition may preserve wisdom, but length of use alone does not prove superiority.

Argument 4: Same Action, Different Punishment

- Conclusion: The rule is not just.
- Stated reason: A just rule should treat the same action the same way, but this rule punishes one student and excuses another for the same action.
- Hidden assumption: Relevant circumstances are truly the same; no morally important difference justifies the different treatment.
- Test: This is a stronger argument if the cases really are alike in all relevant ways. If important circumstances differ, the argument needs more information.

Repair and Original Test Case

- Repaired arguments should either narrow an overclaim or add a reason that directly supports the conclusion.
- Original student arguments should include a clear conclusion and a stated reason. Stronger responses will identify a real hidden assumption rather than repeating the stated reason.

Assessment

Use the repaired argument, original test case, Shakespeare application, portfolio exit paragraph, and bell-puzzle logic test as formative assessment. Strong work should:

- identify conclusions and reasons accurately;
- state hidden assumptions in complete, testable form;
- explain whether the reason actually earns the conclusion;
- improve a weak argument by narrowing the claim or strengthening the support;
- avoid treating impressive language, tradition, honor, or confidence as automatic proof.
- distinguish what logic can test from what must be established by facts, testimony, observation, or moral judgment.

Next Week Preview

Week 2 should move from reasoning in general to **terms**: the words that carry arguments. The next guiding question should be: *Why do arguments become unstable when key words are unclear?* Keep returning to *Macbeth* and *Julius Caesar* so students can practice new logic tools on familiar literary ground.